



JURONG JUNIOR COLLEGE
2018 JC 2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CLASS

18S24

EXAM INDEX

CHEMISTRY

8873/02

Higher 1

Paper 2 Structured Questions

29 August 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams, graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in **Section A (60 marks)** in the space provided on the Question paper.

Answer **one** question in **Section B (20 marks)** in the space provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
Section B	
Lack 3sf in final ans	-1 / NA
Missing/wrong units in final ans	-1 / NA
Total	

This document consists of **20** printed pages.

Section A

Answer **all** the questions in this section, in the space provided.

For
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Use

- 1** An experiment was carried out to determine the percentage of iron in a sample of iron wire.

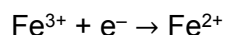
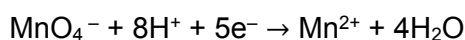
- (a)** A 3.35 g piece of wire was reacted with dilute sulfuric acid, in the absence of air, so that all of the iron atoms were converted to iron(II) ions. The resulting solution was made up to 250 cm³.

- (i)** Write a balanced equation for the reaction between the iron in the wire and the sulfuric acid.

..... [1]

A 25.0 cm³ sample of this solution was acidified and titrated with 0.0250 mol dm⁻³ potassium manganate(VII). 32.0 cm³ of the potassium manganate(VII) solution was required for complete reaction.

The relevant half-equations for the reaction are shown below.



- (ii)** Use these two half-equations to construct an ionic equation for the reaction between manganate(VII) ions and iron(II) ions in acid solution.

.....
.....
..... [2]

- (iii)** Calculate the amount, in moles, of manganate(VII) ions used in the titration.

[1]

- (iv)** Calculate the total amount, in moles, of iron(II) ions in 250 cm³ of the solution.

[2]

- 1 (a) (v) Hence, calculate the percentage by mass of iron in the 3.35 g piece of wire.

[2]

- (b) The electronegativity values of some elements are shown in the table below.

element	electronegativity
aluminium	1.5
chlorine	3.0
iron	1.8

- (i) Use the data to suggest the nature of the bonding in iron(III) chloride. Explain your answer.

.....

[2]

- (ii) When a piece of universal indicator paper is dipped into a solution of iron(III) chloride, the indicator paper turns red.

Suggest an equation to account for the colour change in the indicator paper.

.....

[1]

[Total: 11]

- 2 (a) State the difference between a nanoparticle and a nanomaterial in terms of size.

.....
 [1]

- (b) Many modern sunscreens contain nano-sized particles of titanium dioxide. This substance does **not** absorb ultraviolet radiation.

Suggest how these nano-particles are able to protect skin from ultraviolet radiation.

.....
 [1]

- (c) The engines of modern motor cars have exhaust systems which are fitted with catalytic converters in order to reduce atmospheric pollution from substances such as nitrogen monoxide, NO. These substances are converted to less harmful gases on the catalyst surface before they are released into the atmosphere.

- (i) One of the active materials present in the catalytic converter is platinum nanoparticles. Explain how the use of nanoparticles helps to increase the rate of gaseous reactions in the catalytic converter.

.....

 [2]

- (ii) Give an equation for a reaction involved in the removal of nitrogen monoxide from the exhaust gases of the car, in the catalytic converter.

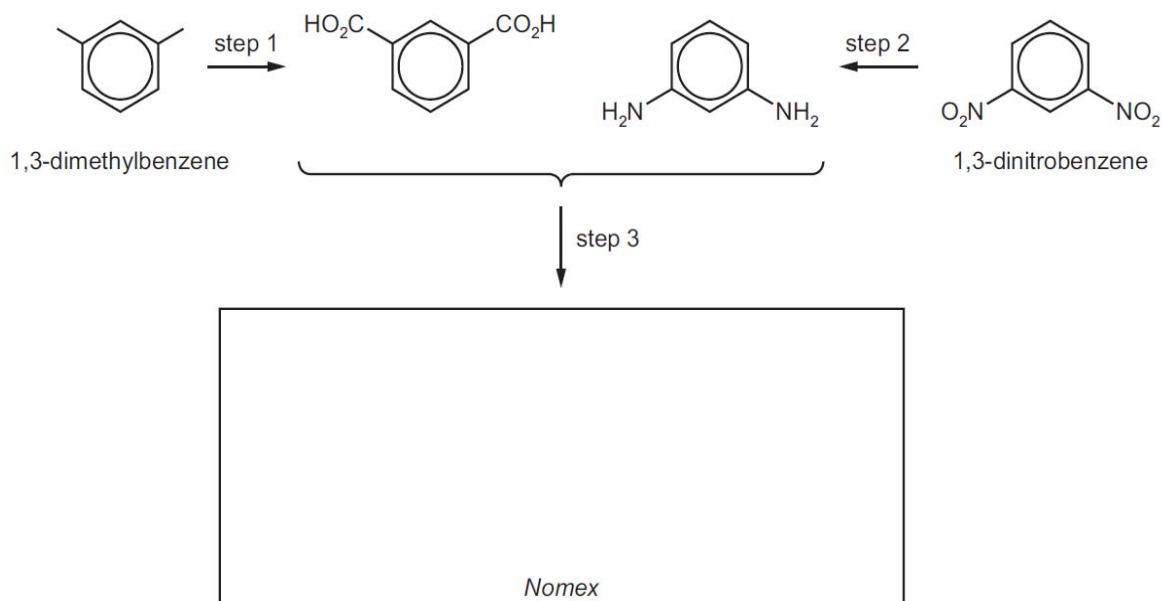
.....
 [1]

- (iii) State one way in which nitrogen monoxide acts as air pollutant.

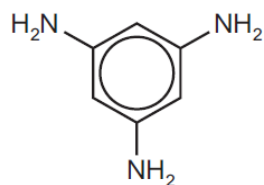
.....
 [1]

[Total: 6]

- 3 1,3-dimethylbenzene is a starting material for the synthesis of the polymer *Nomex*, used in fireproof protective clothing worn by firefighters, military pilots and racing car drivers. The polymer is made from 1,3-dimethylbenzene and 1,3-dinitrobenzene by the following route.



- (a) (i) Draw the structure of one repeat unit of *Nomex* in the box above. [1]
- (ii) State the type of reaction that occurs in step 3. [1]
-
- (iii) Suggest the by-product formed in step 3. [1]
-
- (iv) Suggest how and why the properties of the polymer might change if some of the diamine monomer were replaced with 1,3,5-triaminobenzene.



1,3,5-triaminobenzene

- [1]
-

3 (b) Similar to *Nomex*, Nylon is also a polyamide.

(i) State one use, with reason, of nylon.

.....
.....

[1]

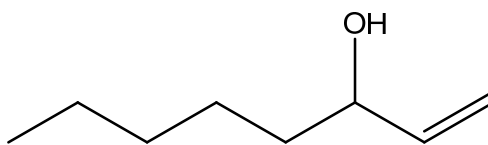
(ii) Nylon is non-biodegradable. What does this mean?

.....
.....

[1]

[Total: 6]

4 1-octen-3-ol has the following structure.



This chemical is found in human sweat and breath, and is thought to attract biting insects such as mosquitoes. It is often used in combination with CO₂ in mosquito traps. It is also found in mushrooms, and sometimes called 'mushroom alcohol'.

- (a) (i) State the type of reaction and what you would observe when 1-octen-3-ol reacts with a solution of bromine in tetrachloromethane (CCl₄).

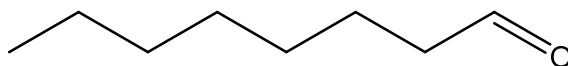
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[2]

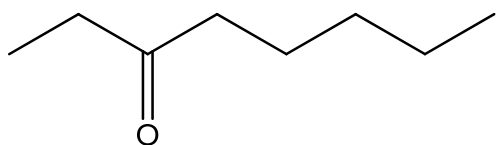
- (ii) Draw the structure of the organic product formed when 1-octen-3-ol reacts with bromine in tetrachloromethane (CCl₄).

[1]

- (b) Other examples of organic compounds containing eight carbon atoms are octanal and octan-3-one.



octanal



octan-3-one

- (i) Octanal and octan-3-one can both be reduced using sodium borohydride, NaBH₄.

Draw the structural formula of the organic product obtained from the reduction of octanal and octan-3-one respectively.

Organic compound	Organic product obtained
Octanal	
Octan-3-one	

[2]

- 4 (b) (ii) Octanal and octan-3-one were heated separately with acidified potassium dichromate(VI). Write an equation for any reaction that occurs.

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.....
.....

[1]

- (c) Oct-3-ene, C_8H_{16} , has two isomers with the same structural formula.
Draw the **displayed** formulae of the two isomers and label them clearly.

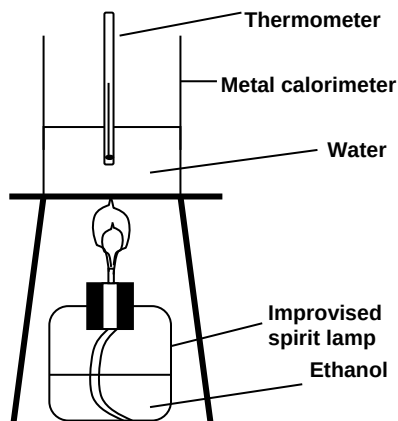
[2]

[Total: 8]

- 5 Biofuels such as ethanol, methanol and methane have been investigated as possible alternative fuels for motor vehicles that currently uses petrol.

(a) The enthalpy change of combustion of ethanol can be determined in the laboratory using the apparatus shown in the diagram below.

The heat produced by the burning fuel warms a known mass of water.



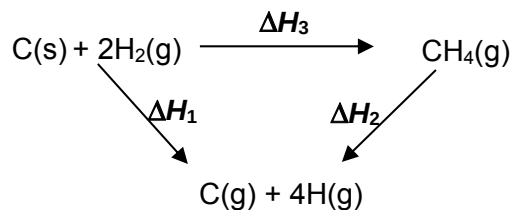
The following results were obtained.

Mass of ethanol burnt ($M_r = 46.0$)	0.10 g
Initial temperature recorded	25.0 °C
Maximum temperature recorded	x °C
Enthalpy change of combustion of ethanol calculated	-840 kJmol ⁻¹

Assume the total heat capacity of the calorimeter and its contents is 120 J K⁻¹, calculate the maximum temperature, x °C, recorded.

[3]

- 5 (b) Consider the energy cycle below involving methane and the enthalpy changes, ΔH_1 , ΔH_2 and ΔH_3 .



- (i) State *Hess's Law*.

.....

[1]

- (ii) Using the data given below, calculate ΔH_1 .



[1]

- (iii) What enthalpy change does ΔH_3 represent?

.....

[1]

- (iii) Using your answer to (b)(ii) and $\Delta H_3 = -75 \text{ kJ mol}^{-1}$, calculate ΔH_2 .

[1]

- (iv) Write an equation to represent the average C-H bond energy in methane, CH_4 .

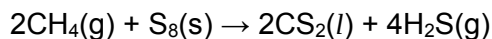
.....

[1]

- (v) Hence, calculate the average C-H bond energy in methane.

[1]

- 5 (c) Methane and sulfur react to produce carbon disulfide (CS_2), a liquid often used in the production of cellophane.



- (i) Write an equation to represent the standard enthalpy change of combustion of liquid carbon disulfide.

..... [1]

The tables below contain values of standard enthalpy change of combustion, ΔH_c , and standard enthalpy change of formation, ΔH_f , of some substances.

Table (I)

	C(graphite)	S(s)
$\Delta H_c / \text{kJ mol}^{-1}$	-394	-297

Table (II)

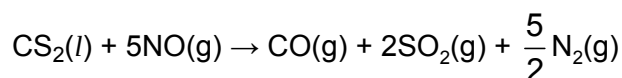
	$\text{CS}_2(\text{l})$	CO(g)
$\Delta H_f / \text{kJ mol}^{-1}$	+88	-111

Use of relevant data given in **Tables (I)** and **(II)** is required to solve **c(ii)** and **c(iii)** of this question.

- (ii) Calculate the standard enthalpy change of combustion of $\text{CS}_2(\text{l})$.

[2]

- (iii) Carbon disulfide reacts explosively with nitrogen monoxide (NO) according to the following equation.



Given the enthalpy change of this reaction is $-1243 \text{ kJ mol}^{-1}$, calculate the enthalpy change of formation of $\text{NO}(\text{g})$ in this reaction.

[2]

[Total: 14]

- 6 Ammonium nitrate is an important fertiliser made by the acid-base reaction between ammonia and nitric acid.

- (a) (i) Write an equation for the production of ammonium nitrate from ammonia and nitric acid.

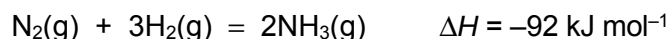
..... [1]

- (ii) Describe the meaning of the terms *Bronsted acid* and *Bronsted base*. Identify the acid and base for the reaction in (a)(i), in terms of the Bronsted-Lowry theory.

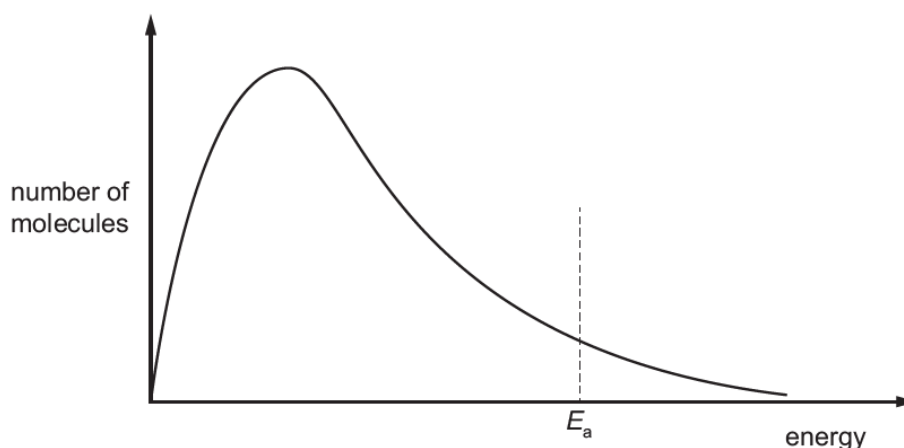
..... [2]
.....

The ammonia for this reaction is produced by the Haber process and the nitric acid is produced by the oxidation of ammonia.

- (b) The Haber process involves a reaction between nitrogen and hydrogen at a temperature of 450 °C and a pressure of 20 000 kPa.



The diagram below shows the Boltzmann distribution curve for the uncatalysed reaction between nitrogen and hydrogen at 450 °C. The *activation energy* of the reaction is labelled as E_a .



- (i) The Haber process is an example of a large-scale gaseous reaction that is catalysed. State the catalyst used.

..... [1]

- (ii) Explain the meaning of the term *activation energy*.

..... [1]
.....
.....

- (iii) On the energy axis of the graph above, clearly mark the position of the activation energy of the catalysed reaction between nitrogen and hydrogen with the letter C.

[1]

- 6 (b) (iv) At a higher reaction temperature, the rate of production of ammonia would be greater. Explain why a higher temperature is **not** used despite the fact that it would increase the rate of production of ammonia.

.....

[2]

- (c) The first stage in the production of nitric acid involves the reaction of ammonia with oxygen to form nitrogen monoxide, NO, and water.

Write an equation for this reaction and use oxidation numbers to show that it is a redox reaction.

.....

[3]

- (d) (i) Draw a dot-and-cross diagram of the ammonium ion. Show the outer electrons only.

[2]

- (ii) State the shape of an ammonium ion and give the H–N–H bond angle.

shape.....

bond angle.....

[2]

[Total: 15]

Section B

Answer **one** question from this section, in the space provided.

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- 7 (a) Sodium, silicon and chlorine are all elements in Period 3 of the Periodic Table. Describe the variation in atomic radius, melting point, electrical conductivity and bonding of sodium, silicon and chlorine.

.....

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.....

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[5]

- (b) Three of the oxides of Period 3 elements are sodium oxide, phosphorus(V) oxide and sulfur trioxide.

(i) Write equations for the reactions of each of these oxides with water.

[3]

(ii) Write an equation for the reaction of each oxide with either sodium hydroxide or hydrochloric acid.

[3]

- 7 (b) (iii) Describe the variation in behaviour shown by these oxides across Period 3 as shown in (b)(ii).

.....
 [1]

- (c) (i) Write a balanced equation to represent the lattice energy of sodium chloride.

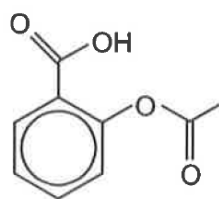
..... [1]

- (ii) Explain why the lattice energy of magnesium oxide is much more exothermic than that of sodium chloride.

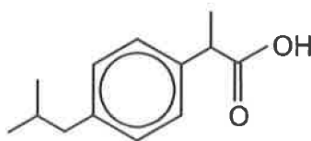
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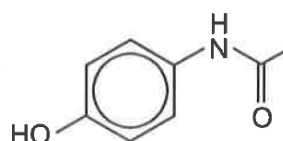
- (d) The pharmaceuticals aspirin, ibuprofen and paracetamol are all painkillers.



aspirin



ibuprofen



paracetamol

Infra-red absorptions can be used to identify functional groups in organic compounds. For example, ethyl ethanoate shows absorptions at $1050\text{--}1330\text{ cm}^{-1}$ and $1710\text{--}1750\text{ cm}^{-1}$ as shown in the *Data Booklet*.

- (i) Identify an infra-red absorption range that will be shown by aspirin and ibuprofen but **not** by paracetamol using the *Data Booklet*.

[1]

- (ii) Identify two of the infra-red absorption range that will be shown by paracetamol but **not** by aspirin or ibuprofen using the *Data Booklet*.

[1]

- 7 (d) (iii) Identify a type of reaction which will occur with both paracetamol and aspirin but **not** with ibuprofen.

..... [1]

- (iv) Write an equation for the reaction of ethanol with ethanoic acid. State the conditions required.

.....
..... [2]

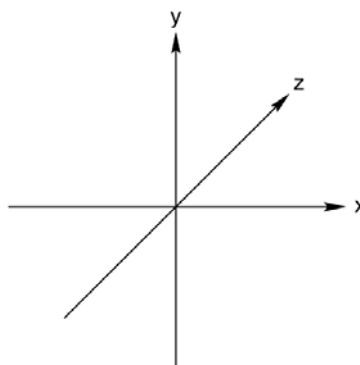
[Total: 20]

8 This question is about phosphorus and its compounds.

- (a) (i) State the full electronic configuration of phosphorus.

..... [1]

- (ii) On the axes below, sketch the shape of the singly-occupied orbital in an atom of phosphorus at ground state.

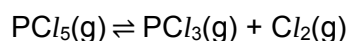


[1]

- (iii) The first ionisation energy of phosphorus is higher than that of sulfur. Explain why.

.....
..... [1]

- (b) The dissociation of phosphorus(V) chloride into phosphorus(III) chloride and chlorine gas is reversible:



The reaction is an example of a *dynamic equilibrium*.

- (i) Explain the meaning of the term *dynamic equilibrium*.

..... [1]

- (ii) In an experiment, 1.00 mol of PCl_5 vapour was heated in a closed 5.00 dm³ flask at 500 K until equilibrium had been established. When equilibrium is established, the percentage conversion of PCl_5 is found to be 51 %. Calculate the equilibrium constant, K_c , at this temperature and state its units.

[2]

- 8 (b) (iii) State and explain what would happen to the equilibrium position if the initial concentration of chlorine was increased.

.....

[1]

- (iv) Draw and name the shapes of PCl_3 and PCl_5 .

[2]

- (v) Explain why PCl_5 sublimes at 160°C while PCl_3 boils at a lower temperature of 76.1°C .

.....

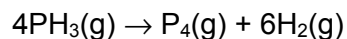
[1]

- (d) Phosphorus compounds are used in the pharmaceutical industries. For example, solutions containing hydrogen phosphates are used as buffers found in the hydrophilic gel used in skin patches. Write equations to show how the solution containing $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$ can act as a buffer when H^+ and OH^- ions are added respectively to the solution.

.....

[2]

- 8 (e) Phosphine, PH_3 , is a widely used fumigant. The thermal decomposition of phosphine into phosphorus and hydrogen is a *first-order reaction*.



- (i) Explain the term in *italics*.

.....
 [1]

- (ii) Sketch a graph of rate against concentration for the above decomposition reaction.

[1]

The half-life of the reaction is 35.0 seconds at 680 °C. Calculate

- (iii) the rate constant for the reaction.

[1]

- (iv) the time taken for 87.5% of the phosphine to decompose.

[1]

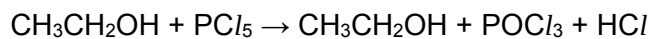
- 8 (f) Phosphorus compounds are also used in organic reactions, as shown in the following reactions. Suggest the type of reaction occurring in each case.

- (i) Phosphoric(V) acid, H_3PO_4 , is used as a catalyst in the reaction.



..... [1]

- (ii) Phosphorus(V) chloride reacts with ethanol.



..... [1]

- (g) Describe, in terms of orbital overlap, the bonding of the two carbon atoms of the C=C bond in $\text{CH}_2=\text{CH}_2$. A clearly labelled diagram may clarify your answer.

[2]

[Total: 20]

